

Ioannis Panitsas

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Research Area

I am a fourth-year Ph.D. candidate in Electrical and Computer Engineering at Yale University, working at the intersection of next-generation cellular networks and applied machine learning.

Research interests: • Cellular Networking • Network Management and Optimization • Machine Learning

Education

Yale University , New Haven, CT, USA Ph.D. Candidate in Electrical and Computer Engineering Advisor: Prof. <i>Leandros Tassiulas</i>	08/2022 – 08/2027
Yale University , New Haven, CT, USA M.Phil. in Electrical and Computer Engineering	08/2022 – 05/2025
Yale University , New Haven, CT, USA M.Sc. in Electrical and Computer Engineering	08/2022 – 05/2024
School of Military Engineering , Loutraki, Greece 2nd Lieutenant, Engineering Corps, Hellenic Army	03/2021 – 05/2022
University of Patras , Patras, Greece B.Sc. & M.Sc. in Electrical and Computer Engineering Advisor: Prof. Dimitrios Serpanos	09/2016 – 07/2021

Publications

- **I. Panitsas** and L. Tassiulas, "TimeRAN: Multi-Task Time-Series Learning for Next-Generation Radio Access Networks," under review.
- **I. Panitsas**, P. Gajjar, E. Ojo, V. K. Shah, and L. Tassiulas, "PRISMA: Multivariate Anomaly Detection and Root-Cause Diagnosis for Open RAN," under review.
- **I. Panitsas**, T. Atalay, D. Stojadinovic, A. Stavrou, and L. Tassiulas, "SlicePilot: Demystifying Network Slice Placement in Heterogeneous Cloud Infrastructures," accepted to *IEEE International Conference on Computer Communications (INFOCOM)*, 2026.
- **I. Panitsas**, I. Ofeidis, and L. Tassiulas, "FedJam: Multimodal Federated Learning Framework for Jamming Detection," accepted to *IEEE International Conference on Computer Communications (INFOCOM)*, 2026.
- **I. Panitsas**, T. Atalay, D. Stojadinovic, A. Stavrou, and L. Tassiulas, "5GC-Bench: A Framework for Stress-Testing and Benchmarking 5G Core VNFs," accepted to *IEEE Wireless Communications and Networking Conference (WCNC)*, 2026.
- **I. Panitsas**, A. Mudvari, A. Maatouk, and L. Tassiulas, "A Deep and Transfer Learning Approach for Handover Management in O-RAN," accepted to *IEEE Wireless Communications and Networking Conference (WCNC)*, 2026.
- **I. Panitsas**, Y. Yigit, L. Tassiulas, L. Maglaras, and B. Canberk, "JamShield: A Machine Learning Detection System for Over-the-Air Jamming Attacks," *Proceedings of the IEEE International Conference on Communications (ICC)*, 2025.
- **I. Panitsas**, A. Mudvari, and L. Tassiulas, "D2Q Synchronizer: Distributed SDN Synchronization for Time-Sensitive Applications," *Proceedings of the IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, 2025.
- Y. Yigit, **I. Panitsas**, L. Maglaras, L. Tassiulas, and B. Canberk, "Cyber-twin: digital twin-boosted autonomous attack detection for vehicular ad-hoc networks," *Proceedings of the IEEE International Conference on Communications (ICC)*, 2024.
- A. Feng, A. Varvarigos, **I. Panitsas**, D. Fernandez, J. Wei, Y. Guo, J. Chen, A. Maatouk, L. Tassiulas, and R. Ying, "TelecomTS: A Multi-Modal Observability Dataset for Time Series and Language Analysis," accepted to International Conference on Machine Learning (ICML).

Industry Experience

- Research Intern**, AT&T Labs – Bedminster, NJ, USA 06/2026 – 08/2026
- Worked within the RAN Technology Research Group on conflict detection and mitigation algorithms enabling concurrent operation of Open RAN applications without performance degradation.
- Wireless Systems Research Intern**, Nokia Bell Labs – Murray Hill, NJ, USA 01/2026 – 05/2026
- Worked within the RAN Platforms Research group on next-generation AI-RAN architectures and neural receiver implementations for improving uplink 5G performance.
- Research Scientist Intern**, A2 Labs – Arlington, VA, USA 07/2025 – 08/2025
- Designed a resource orchestrator for efficient slice placement across cloud infrastructure and developed *5GC-Bench*, a benchmarking framework for testing and profiling 5G Core VNFs.
- Research Scientist Intern**, Kryptowire Labs – Arlington, VA, USA 07/2024 – 08/2024
- Developed *SlicePilot*, a reinforcement learning-based integrated system for optimal network slice placement and resource allocation.
- Embedded Software Engineer Intern**, Renesas Electronics – Patras, Greece 08/2020 – 10/2020
- Performed link-layer penetration testing and security evaluation of the Bluetooth Low Energy 5.0 protocol, identifying protocol-level vulnerabilities through crafted over-the-air exploits.

Teaching Experience

- Teaching Assistant**, Yale University – New Haven, CT, USA 01/2025 – 05/2025
- Course: Introduction to Computing for Engineers and Scientists, ENAS 130
- Teaching Assistant**, Yale University – New Haven, CT, USA 08/2024 – 12/2024
- Course: Introduction to Communications and Control, EENG 202
- Teaching Assistant**, Yale University – New Haven, CT, USA 01/2024 – 05/2024
- Course: Introduction to Computing for Engineers and Scientists, ENAS 130

Workshops

- **STAIRWAI Workshop on AI for Wireless 2026**, Rutgers University, New Brunswick, NJ, USA
Presented a poster on *SlicePilot*, a reinforcement learning-based framework for optimal network slice placement.
- **CyberPowder Fellows Program 2025**, University of Utah, Salt Lake City, UT, USA
Participated in an on-site research workshop on the POWDER wireless testbed.
- **Open AI Cellular Workshop 2024**, Mississippi State University, Starkville, MS, USA
Engaged in hands-on experimentation at the OAIC research workshop.
- **Young Gladiators Workshop 2024**, Northeastern University, Boston, MA, USA
Participated in an on-site research workshop on the Colosseum wireless emulator.

Reviewer Roles

Journals: IEEE Communications Magazine, IEEE/ACM Transactions on Networking, IEEE Transactions on Cognitive Communications and Networking, IEEE Transactions on Information Forensics and Security, IEEE Wireless Communications Letters

Conferences: ACM SIGMETRICS, IEEE Dynamic Spectrum Access Networks (DySPAN), IEEE Military Communications Conference (MILCOM), IEEE Vehicular Technology Conference (VTC)

Skills

- **Programming:** Python, C/C++ , OpenMP, JavaScript, Java, SQL, MongoDB, HTML, CSS, Node.js
- **Tools:** Docker, Kubernetes, Helm
- **Wireless/5G Systems:** OpenAirInterface (OAI), srsRAN, Open5GS, NVIDIA Aerial, Sionna-RK, Sionna-RT

Languages

- English (fluent), Greek (native)